Adway S. Wadekar

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EDUCATION

Ph.D. in Statistics, University of Michigan, Ann Arbor, expected 2030.
Advisor: Prof. Jake Soloff

B.S. in Mathematics (with distinction) and Statistical Science (with high distinction), Duke University, 2025.
Minor in Economics.
Advisors: Prof. Ezra Miller (mathematics) and Prof. Jerry Reiter (statistical science).

RESEARCH INTERESTS

Foundations of modern data science: shape constraints, empirical Bayes, graphical models, nonparametrics, AI-assisted statistics.

Population and statistical genetics: ancestral recombination graphs, evolutionary basis of trait genetics, demographic inference.

High-performance and scientific computing, optimization, and intersections with statistics.

Applications of statistics to the social sciences, including sociology and data/computational journalism.

RESEARCH EXPERIENCE

Graduate Research

2026– Research Assistant, Department of Statistics, University of Michigan
Advisor: Prof. Jake Soloff

Large-scale inference and testing methodology adapted to modern data science using techniques from empirical Bayes, statistical machine learning and shape-constrained estimation.

2026– Research Assistant, Department of Statistics, University of Michigan
Advisor: Prof. F. Richard Guo

Non-parametric testing for structure in graphical models in the “hunt-and-test” framework. Embedding machine learning into scoring procedures to search for plausible alternative directions, resulting in increased power.

2025– Research Assistant, Department of Statistics, University of Michigan
Advisor: Prof. Jonathan Terhorst

Methods to improve computational feasibility of ancestral recombination graph inference using techniques from neural particle filtering.

Selected Undergraduate Research

2024–25 Research Assistant, Department of Biomedical Informatics, Harvard Medical School
Advisor: Prof. Chirag Patel

Developed methods to translate estimators for polygenic risk scores to exposomic and proteomic data. Supported through the Harvard Summer Institute in Biomedical Informatics.

2022–25 Research Assistant, Department of Statistical Science, Duke University
Advisor: Prof. Jerry Reiter

Developed a technique to incorporate complex survey design into estimates of population-level accuracy for machine learning classifiers trained and evaluated on survey data. Developed a decision-theoretic sensitivity analysis framework that accounts for data uncertainties common to the social sciences.

PUBLICATIONS

Journal Articles

1. Wadekar AS and Reiter JP, Evaluating binary outcome classifiers estimated from survey data, *Epidemiology*, vol. 35, no. 6, pp. 805–812, 2024. DOI: 10.1097/EDE.0000000000001776.
2. Wadekar AS, Understanding opioid use disorder using tree-based classifiers, *Drug and Alcohol Dependence*, vol. 208, p. 107 839, 2020. DOI: 10.1016/j.drugalcdep.2020.107839.

Conference Proceedings

1. Wadekar AS, A psychosocial approach to predicting substance use disorder (sud) among adolescents, in *Proceedings of IEEE/ACM Intl. Conference on Social Networks Analysis and Mining*, 2020, pp. 819–826. DOI: 10.1109/ASONAM49781.2020.9381378.

Under Review

1. Wadekar AS and Soloff JA, Calibration without labels in multiple testing, submitted to NeurIPS, 2026. DOI: https://doi.org/10.48550/arXiv.2606.19737.
2. Wadekar AS and Reiter JP, Assessing decision-making confidence in the presence of data uncertainty, major revision at *Statistics and Public Policy*, 2025. DOI: https://doi.org/10.48550/arXiv.2504.17043.

PRESENTATIONS

May 2023 "Truth or consequences? A principled path to evaluating classifiers using survey data," Symposium on Data Science and Statistics, St. Louis, Missouri.

Jan. 2023 "Using weights to improve the reliability of classification metrics with complex survey data," Dept. of Statistical Science, Duke University.

July 2019 "Predicting Opioid Use Disorder Using a Random Forest," IEEE 43rd Computer Software and Applications Conference, Milwaukee, Wisconsin (peer-reviewed fast abstract).

Mar. 2018 "Grade-level Participation in the Advanced Placement Curriculum," IEEE Integrated STEM Education Conference, Princeton, New Jersey.

AWARDS, HONORS, AND RESEARCH SUPPORT

National-level awards

2026 Department of Energy Computational Science Graduate Research Fellowship.

2026 Honorable Mention (Directorate for Mathematical Sciences), National Science Foundation Graduate Research Fellowship.

2025 Honorable Mention (Directorate for Computer and Information Science and Engineering), National Science Foundation Graduate Research Fellowship.

2024 Summer Institute in Biomedical Informatics (SIBMI) Research Fellowship, Harvard University ($4,500).

University-level awards

2026	Michigan Institute for Computational Discovery and Engineering Fellowship, University of Michigan ($4,500).
2025	Hubert M. Blalock Fellowship, Center for Statistics and the Social Sciences, University of Washington, Seattle (declined, $5,000).
2025	Genome Science Training Program Fellowship (declined for affiliate status), University of Michigan.
2024	Nominee for Churchill Scholarship (2 endorsed each year), Duke University.
2024	Semifinalist (7 semifinalists, 3 awardees out of 1700+ undergraduates) for Faculty Scholars Award, Duke University.
2024	Program for Research for Undergraduates (PRUV) Fellowship, Dept. of Mathematics, Duke University ($3,500).
2023, 2024	Deans' Summer Research Fellowship, Duke University (declined, $3000).
2023	Conference Grant, Office of Undergraduate Research Support, Duke University ($500).
2022	Huang Research Fellowship, Initiative for Science and Society, Duke University ($6,000).

Miscellaneous awards

2025	Fifth Place for Breaking News Story of the Year, Associated College Press.
2023	Finalist submission for Fischer-Zernin Award for Local Journalism, Duke University.
2023	Finalist submission for Melcher Award for Excellence in Student Journalism, Duke University.
2021	Scholar Award, Regeneron Science Talent Search.
2019	Certificate of Honorable Mention (administered by the American Statistical Association), Intel International Science and Engineering Fair.

TEACHING EXPERIENCE

University of Michigan

Graduate Student Instructor for STATS 250, Introduction to Statistics and Data Analysis, Fall 2025, Winter 2026.

Duke University

TA for STA 602L, Bayesian Statistics (master's level), Spring 2025.

TA for STA 432, Theory and Methods of Statistical Learning and Inference, Fall 2024

TA for DECISION 618, Data Analytics for Business (Fuqua School of Business), Fall 2021-23.

Grader for MATH 340, Adv. Introduction to Probability, Spring 2025.

Grader for MATH 230, Probability, Fall 2022, 24.

Grader for MATH 112L, Spring 2022, 23.

Help room TA for MATH 122L, Introductory Calculus II, Spring 2022.

SERVICE

Referee Service

Statistical Science, NeurIPS, Journal of the American Medical Informatics Association, Drug and Alcohol Dependence.

University Service

At-Large Member for Senior Class Gift Committee, Duke University.

Trinity Ambassador for the Department of Statistical Science, Duke University.

Student application reviewer for Huang Fellows Program, Initiative for Science and Society, Duke University.

Blue Devil Buddies peer mentor, Duke University.

OTHER POSITIONS

2021–25	Creative Intern, Duke Men's Basketball.
2021–24	News Editor (2023-24), University News Editor (2022-23) and Staff Reporter (2021-22), Duke Chronicle.
2019–20	Photographer-in-Residence, Town of Westborough, Massachusetts.

SOFTWARE

Proficient in R, Python, assorted bioinformatics tools (PLINK, VCF/BCFtools, etc.).

Updated August 2026